

Supplementary Table S1. Variable descriptions and encoding schemes.

Variable	Description	Encoding
Age	Age at diagnosis	years / 100
Menopausal status	Pre-menopausal or post-menopausal status at the time of diagnosis	post = 1 pre/peri = 0
Histological grade	Tumor differentiation grade (Nottingham system)	well = 1 moderate = 2 poor = 3
ER status	Estrogen receptor status assessed by immunohistochemistry	positive = 1 negative = 0
PR status	Progesterone receptor status assessed by immunohistochemistry	positive = 1 negative = 0
HER2 status	Human epidermal growth factor receptor 2 status assessed by immunohistochemistry	positive = 1 negative = 0
TMB	Total number of somatic mutations per sample	Continuous: rank-based inverse normal transform
FGA	Fraction of genome altered	Continuous: (0–1)
TP53 SNVs	Somatic single nucleotide variant counts in TP53	total SNV count
PIK3CA SNVs	Somatic single nucleotide variant counts in PIK3CA	
TP53 CADD	Maximum combined annotation dependent depletion pathogenicity score of TP53 variants per sample	maximum CADD score per sample
PIK3CA CADD	Maximum combined annotation dependent depletion pathogenicity score of PIK3CA variants per sample	maximum CADD score per sample
CCND1 CNAs	Copy number alteration level of CCND1	Continuous: raw copy number value
MDM4 CNAs	Copy number alteration level of MDM4	Continuous: raw copy number value

Supplementary Table S2. Missing value rates before imputation across the MSK-IMPACT and METABRIC cohorts. Values are reported as the number and percentage of missing samples for each clinical and molecular variable prior to k-nearest neighbor (kNN) imputation.

Variable	MSK-IMPACT (%)	METABRIC (%)
Menopausal Status	7/645 (1.1%)	69/503 (13.7%)
Histological Grade	104/645 (16.1%)	18/503 (3.6%)
ER Status	4/645 (0.6%)	4/503 (0.8%)
PR Status	7/645 (1.1%)	69/503 (13.7%)
HER2 Status	33/645 (5.1%)	69/503 (13.7%)
FGA	7/645 (1.1%)	48/503 (9.5%)

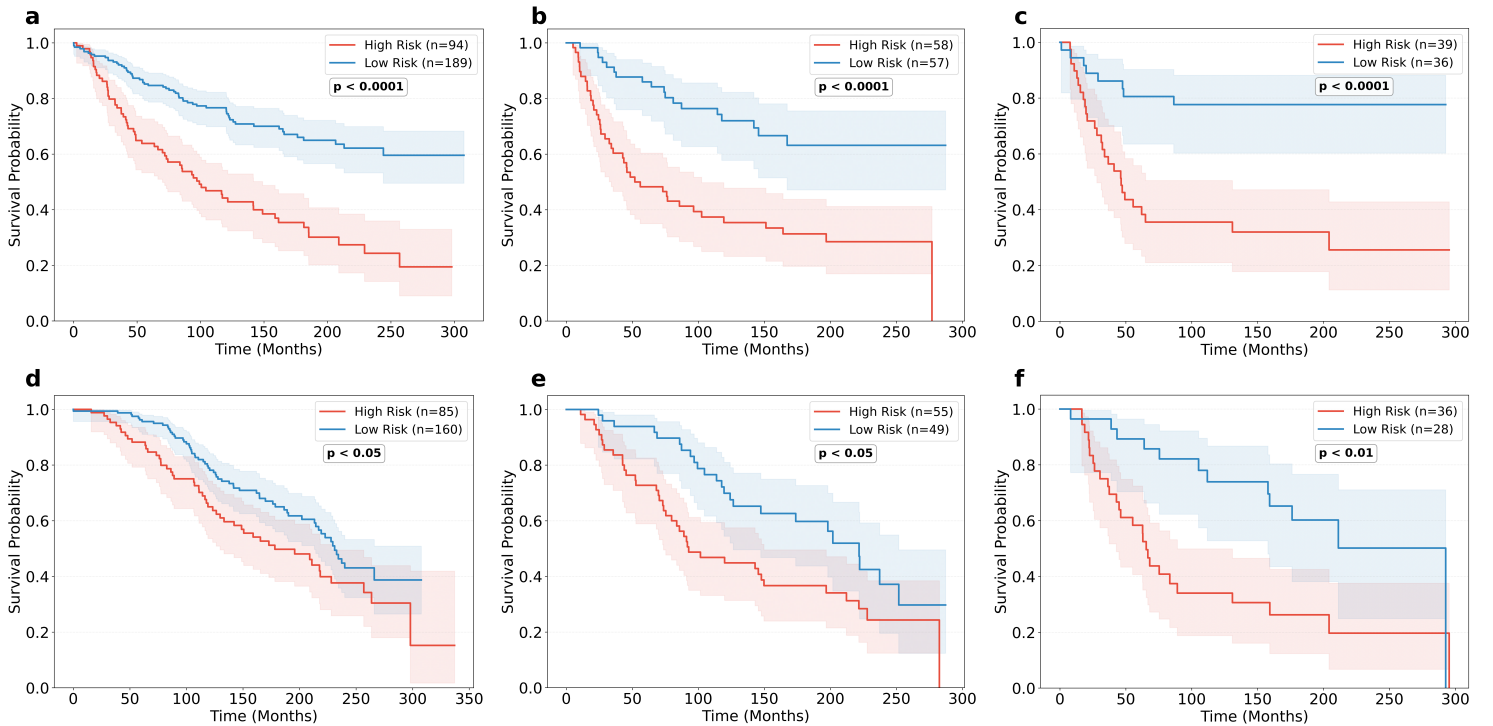
Supplementary Table S3. Optimal hyperparameters for each model under each experimental setting. MSK: MSK-IMPACT cohort; META: METABRIC cohort. Hyperparameters for cross-cohort validations were selected via inner cross-validation on the training cohort only. The final model hyperparameters were determined as the mode of optimal configurations across 25 outer cross-validation runs on the training combined cohort.

Experiment	Model	Hyperparameters
Train MSK → Test META	RF	n_estimators = 100, max_depth = 4, min_samples_leaf = 3, class_weight = 'balanced_subsample', criterion = 'entropy'.
	GBM	n_estimators = 100, max_depth = 3, learning_rate = 0.05, subsample = 0.7, criterion = 'squared_error', loss = 'log_loss'.
	AdaBoost	n_estimators = 300, max_depth = 3, learning_rate = 0.05.
	XGBoost	n_estimators = 200, max_depth = 3, learning_rate = 0.05, subsample = 0.9, eval_metric = 'logloss'.
Train META → Test MSK	RF	n_estimators = 200, max_depth = 4, min_samples_leaf = 3, class_weight = 'balanced_subsample', criterion = 'gini'.
	GBM	n_estimators = 100, max_depth = 4, learning_rate = 0.03, subsample = 0.5, criterion = 'squared_error', loss = 'exponential'.
	AdaBoost	n_estimators = 200, max_depth = 3, learning_rate = 0.01.
	XGBoost	n_estimators = 300, max_depth = 4, learning_rate = 0.01, subsample = 0.7, eval_metric = 'logloss'.
Combined Cohort (The Final Model)	RF	n_estimators = 300, max_depth = 4, min_samples_leaf = 7, class_weight = 'balanced', criterion = 'entropy'.
	GBM	n_estimators = 300, max_depth = 3, learning_rate = 0.01, subsample = 0.5, criterion = 'squared_error', loss = 'exponential'.
	AdaBoost	n_estimators = 200, max_depth = 3, learning_rate = 0.05.
	XGBoost	n_estimators = 300, max_depth = 3, learning_rate = 0.01, subsample = 0.5, eval_metric = 'logloss'.

Supplementary Table S4. Predictive performance of machine learning models for breast cancer relapse prediction across experimental settings. Metrics include area under the receiver operating characteristic curve (AUC), area under the precision–recall curve (AUCPR), recall, precision, F1-score, and specificity. For the combined cohort setting, results are reported as mean $\pm$ standard deviation across 5-fold cross-validation repeated five times (25 runs). MSK: MSK-IMPACT cohort; META: METABRIC cohort.

Model	Experiments	AUC	AUCPR	Recall	Precision	F1	Specificity
RF	Train MSK $\rightarrow$ Test META	0.714	0.663	0.720	0.616	0.664	0.616
	Train META $\rightarrow$ Test MSK	0.741	0.689	0.722	0.669	0.694	0.657
	Combined cohort (5-fold)	0.781 $\pm$ 0.024	0.758 $\pm$ 0.029	0.721 $\pm$ 0.053	0.693 $\pm$ 0.030	0.706 $\pm$ 0.033	0.707 $\pm$ 0.042
GBM	Train MSK $\rightarrow$ Test META	0.669	0.603	0.634	0.579	0.605	0.605
	Train META $\rightarrow$ Test MSK	0.700	0.646	0.604	0.645	0.624	0.681
	Combined cohort (5-fold)	0.785 $\pm$ 0.025	0.775 $\pm$ 0.029	0.693 $\pm$ 0.055	0.711 $\pm$ 0.033	0.701 $\pm$ 0.035	0.741 $\pm$ 0.043
AdaBoost	Train MSK $\rightarrow$ Test META	0.674	0.627	0.625	0.585	0.604	0.620
	Train META $\rightarrow$ Test MSK	0.717	0.675	0.551	0.642	0.593	0.705
	Combined cohort (5-fold)	0.784 $\pm$ 0.026	0.772 $\pm$ 0.031	0.692 $\pm$ 0.062	0.703 $\pm$ 0.034	0.696 $\pm$ 0.037	0.732 $\pm$ 0.047
XGBoost	Train MSK $\rightarrow$ Test META	0.668	0.631	0.578	0.604	0.590	0.675
	Train META $\rightarrow$ Test MSK	0.688	0.631	0.522	0.632	0.572	0.708
	Combined cohort	0.788 $\pm$ 0.026	0.778 $\pm$ 0.028	0.691 $\pm$ 0.059	0.711 $\pm$ 0.039	0.700 $\pm$ 0.039	0.742 $\pm$ 0.048

Supplementary Fig. S1. Subtype-stratified Kaplan-Meier survival curves. Disease-free survival (DFS; **a–c**) and overall survival (OS; **d–f**) for Luminal A, Luminal B, and Triple-Negative subtypes, respectively. Risk groups were defined using a probability threshold of 0.5 based on out-of-fold predictions from the combined cohort cross-validation. Shaded regions represent 95% confidence intervals.



Supplementary Fig. S2. Decision plots illustrating model-derived relapse risk predictions for two representative patients: **a** high-risk patient prediction. **b** low-risk patient prediction. Each horizontal line represents the cumulative contribution of a feature to the final risk probability, starting from the baseline of 0.5. Red segments indicate features pushing predictions toward relapse (positive SHAP values) and blue segments indicate features pushing predictions toward non-relapse (negative SHAP values). Feature values for each patient are shown on the left. The final prediction $f(x)$ is indicated by the vertical line.

